Your Tech Solution

YT SOLUTION

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From Sensing to Control

Your Tech Solution
YT SOLUTION

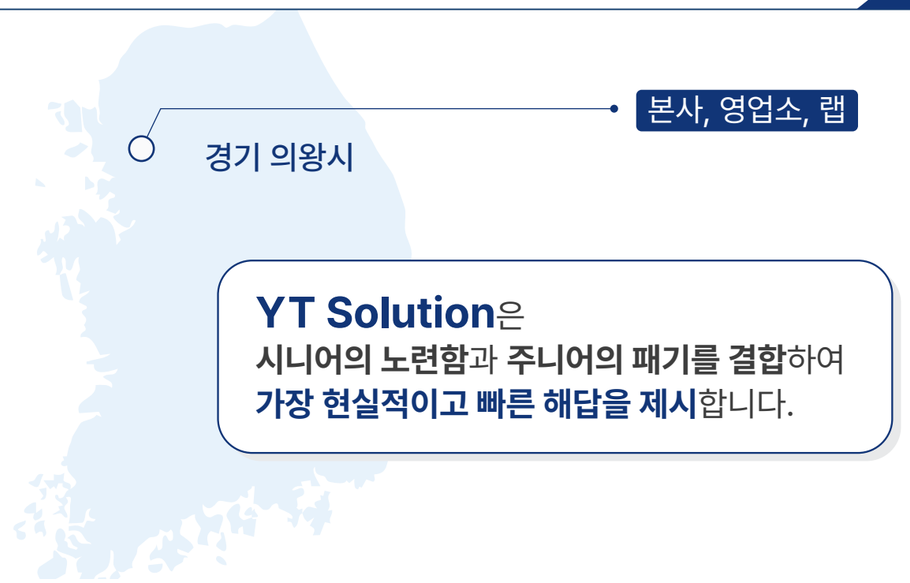
사업자명 (주)와이티솔루션

사업영역 Machine vision solution
Special sensors
PLC component & system

임직원수 4명

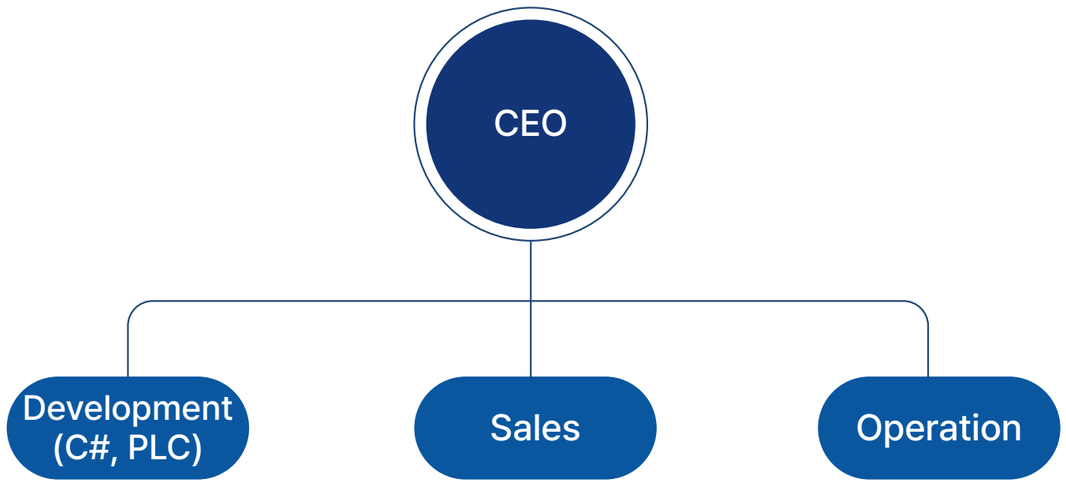
위치 본사 : 경기 의왕시

회사위치



YT Solution은
시니어의 노련함과 **주니어의 패기를 결합**하여
가장 현실적이고 빠른 해답을 제시합니다.

조직도



WHY YT Solution? (원스톱 엔지니어링)

Core Business Domains

1 공식 총판/파트너십
본사 직소통을 통한 납기·단가 경쟁력 및 최신 기술 지원

2 자체 랩 기반 PoC 지원
샘플 테스트, 렌즈/조명 광학 시뮬레이션 및 평가 리포트 제공

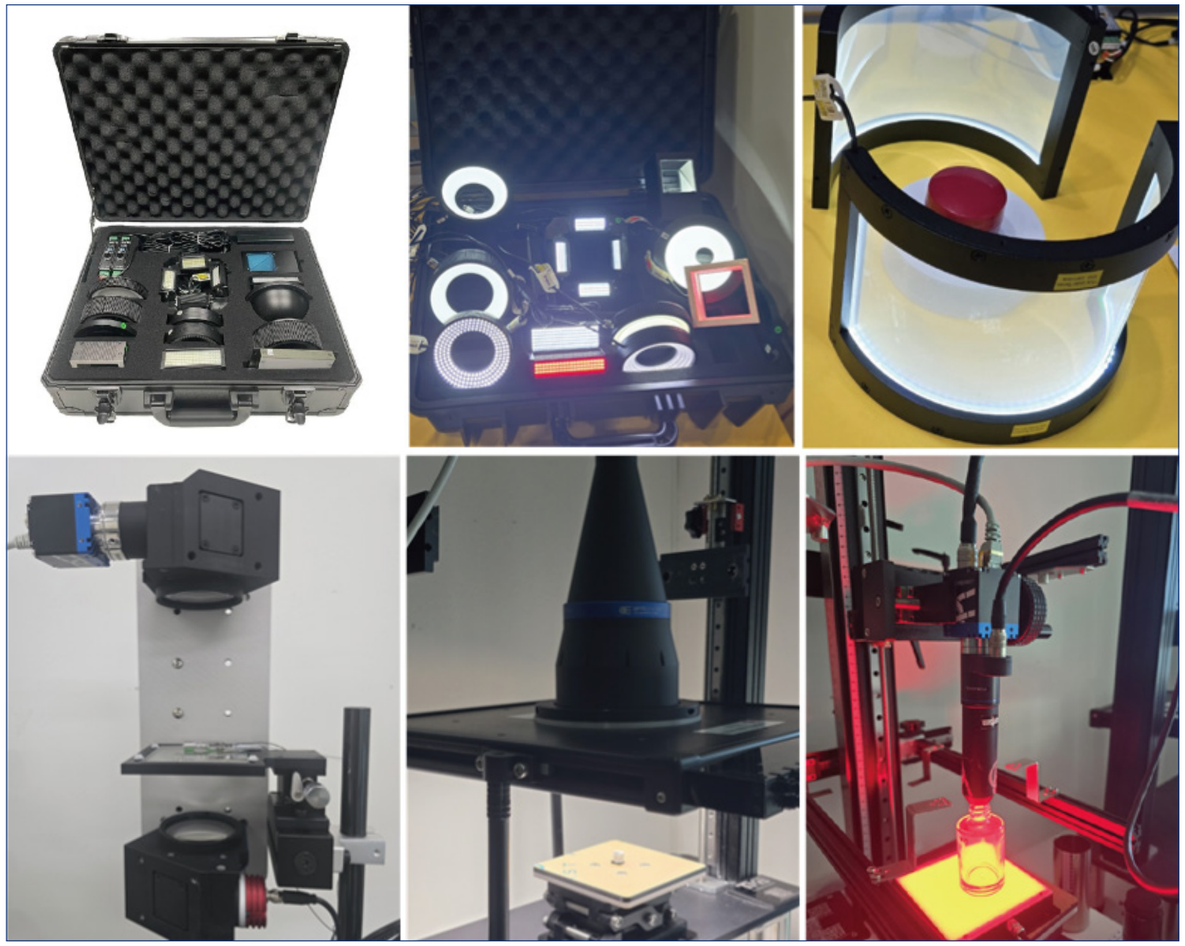
3 원스톱 엔지니어링
PoC(사전 검증) → 광학/제어 BOM 최적화 → 제품 공급
→ 셋업 및 기술 지원

주요 레퍼런스 Main Reference

Vision Solution Demo Lab & Testing Kit

- **Vision solution Process**

1. 의뢰
2. 샘플 수령
3. 조명/렌즈 매칭
4. 결과 리포트 및 견적

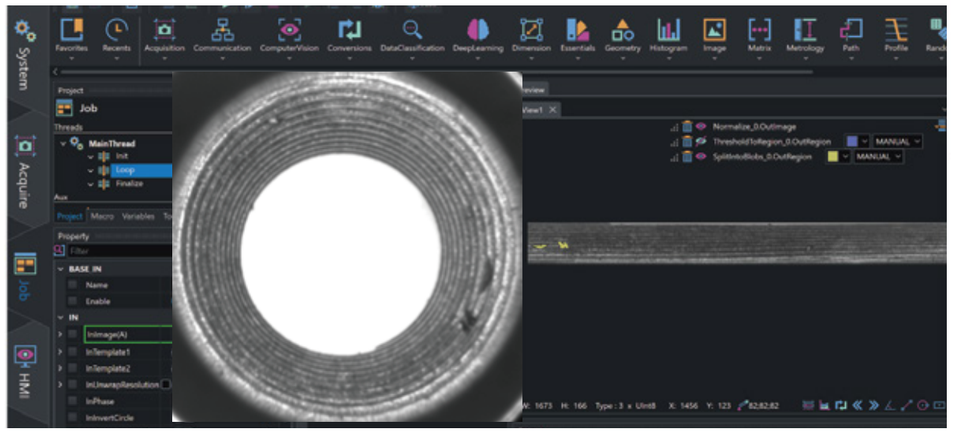


Vision Solution Authorized distributorship

OPTO ENGINEERING	TMS LITE	TPL VISION Active LED solutions
• 360도 내/외부 결함 검사 • 텔레센트릭 계측 및 결함 검사 • 리퀴드 솔루션	• Visible/IR/SWIR 조명 • 랩/현장 조명 데모키트 • 조명 컨트롤러 • 테스트 스테이션	• 라인스캔, 초고속 바 조명 • High Power 링조명 (Angle changer) • 초고속 | 초고휘도 라인스캔 특수 조명(철도 레일 검사용)







1 · SYSTEM

- Configure devices and hardware
- Manage multi-user project access
- Support industrial communication protocols & hardware

2 · ACQUIRE

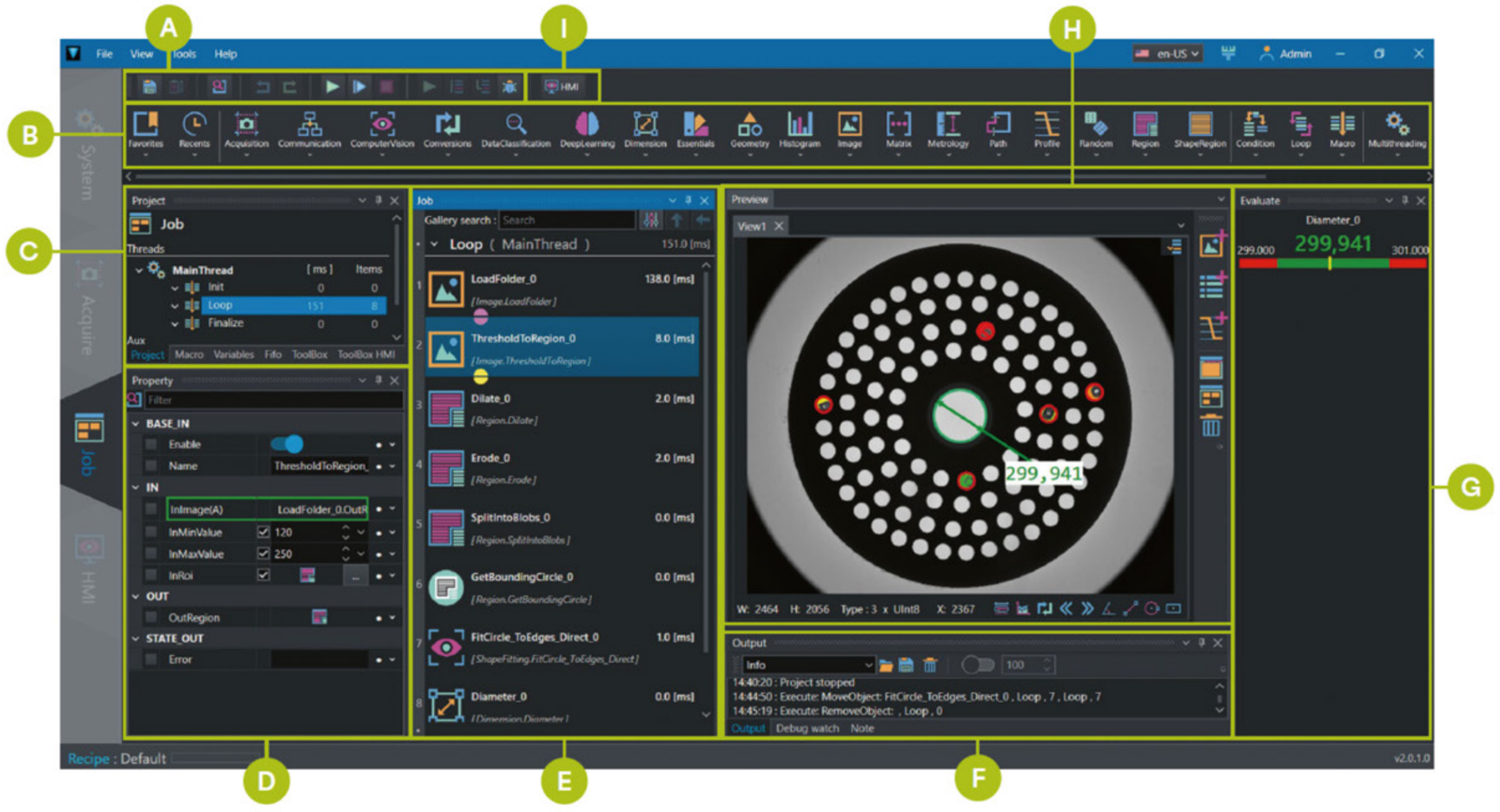
- Multi-camera image acquisition
- Live image preview and monitoring
- Full access to GenICam settings
- Supports any GenTL-compliant camera & major third-party camera SDKs

3 · JOB & HMI DESIGN

- Vision and HMI design in one platform
- Drag & drop interface builder
- Direct vision tool-to-HMI linking
- Advanced UI customization

4 · HMI EXECUTION

- Deploy custom operator interfaces
- Display live inspection results
- Monitor and control applications
- Modern and intuitive user experience



A EXECUTION PANEL

Save, run in a loop or once, stop and debug your project. Select whether to run only the main thread or multiple threads simultaneously.

B JOB TOOL BAR

Tools are the bricks of an OEVIS® project. Each tool is composed of an action that is performed when the tool is run and a list of properties that customize its behavior.

C PROJECT EXPLORER

View Project parts (init, loop and finalize), manage Macros, create global Variables and Fifo queues, view and search tools.

D PROPERTY INSPECTOR

Display and set all the tools properties, including name, input & output values. Parameters followed by the "(A)" symbol automatically link to the first compatible value in the execution flow (if existing).

E JOB PANEL & SEARCH BAR

This panel shows the list of selected tools. The connection between tools is shown by half-circles located at the top (inputs) or at the bottom (outputs) of each tool. Different colors identify connections between different tools.

F DEBUG TOOLBOX

Display the event log, keep track of the property value of the selected tools when the program is running and test the connection to a Modbus server.

G EVALUATE

Visually display dimensional values or any other numerical property to quickly verify if it falls within a predefined range.

H VISUALIZER

Preview the results of image analysis and display edges or other properties via drag & drop functionality from the property panel. Add multiple views, display data tables and much more.

I HMI DESIGN

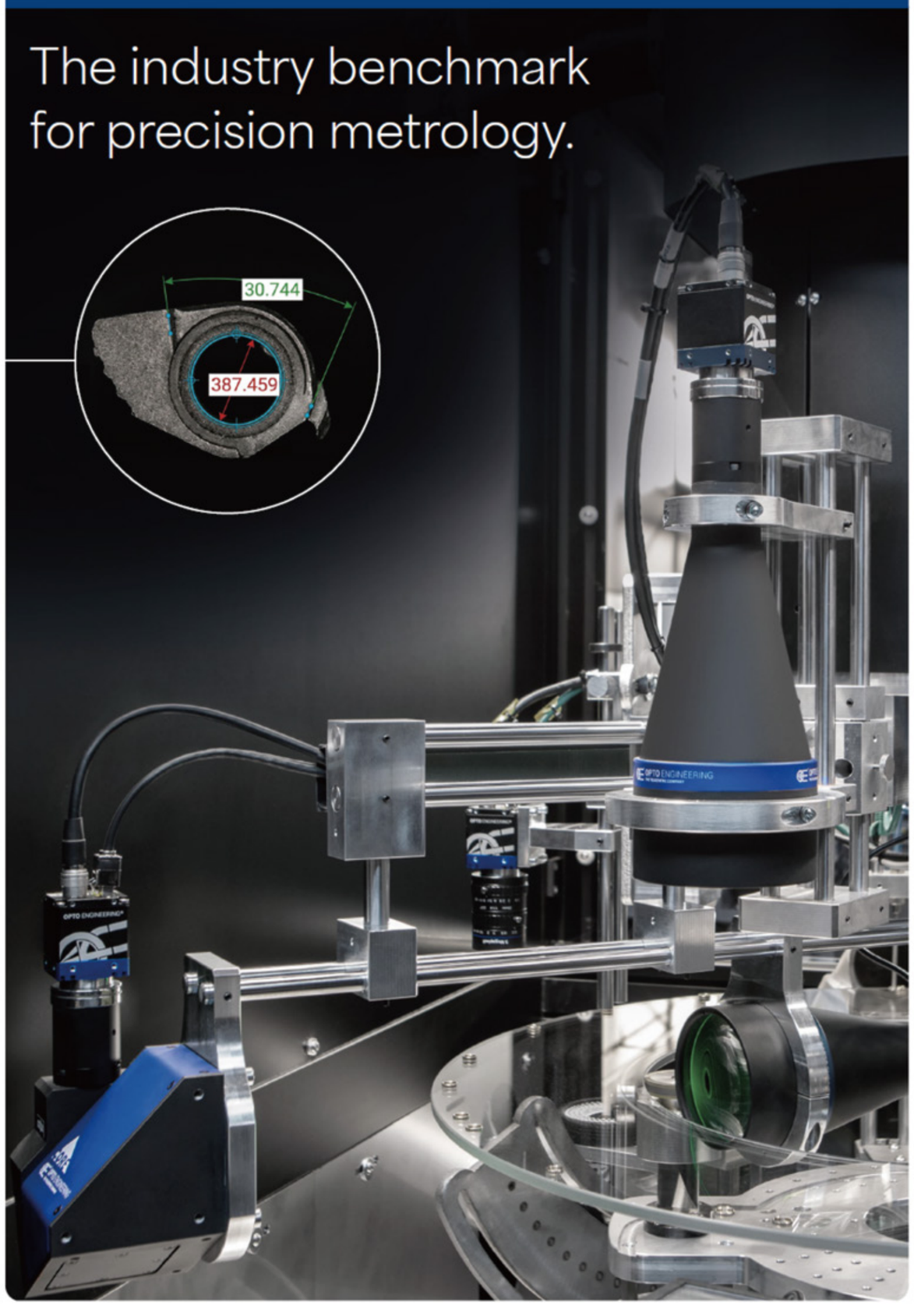
Design and customize your application's front end with a built-in advanced graphic editor.

Supported IPCs: Advantech · ADLINK · IMAGO Technologies · Neousys
Additional hardware platforms available on request.

Industrial connectivity: PROFINET[1] · EtherCAT[1] · EtherNet/IP[1] · OPC UA Client[2] · Modbus TCP[1] · Modbus Client/Server TCP/RTU[2] · MQTT Client[2] · HTTP[2] · WebSocket[2] · RS232 / RS485[2] · TCP/IP[2]
[1] Via Hilscher CIFX cards, [2] Native support. Additional protocols available on request.

TELECENTRIC LENSES

The industry benchmark for precision metrology.



TELECENTRIC LENSES • Wide range

CLASSIC

Telecentric and bi-telecentric lenses



- From area-scan to line-scan sensors
- High telecentricity, consistent results
- Industrial design with fixed aperture

COMPACT

Ultra-compact telecentric and bi-telecentric lenses



- Up to 70% smaller than conventional telecentric lenses
- High-end optical performance, compact design

FOCUS TUNABLE

Telecentric lenses with integrated liquid lens technology



- Remote focus adjustment in milliseconds
- Inspection & measurement across multiple working distances

COAXIAL

Coaxial telecentric lenses



- Built-in coaxial light for reflective parts
- High numerical aperture, short exposure times

FLAT

Flat telecentric lenses



- Built for line-scan cameras
- Flat, compact design for tight spaces

MULTI-MAG

Dual magnification telecentric lenses



- Dual magnification, one lens
- Precise measurements with no recalibration required

3D

Bi-telecentric lenses with Scheimpflug adjustment



- Tilt adjustment for oblique measurements
- Radially undistorted images

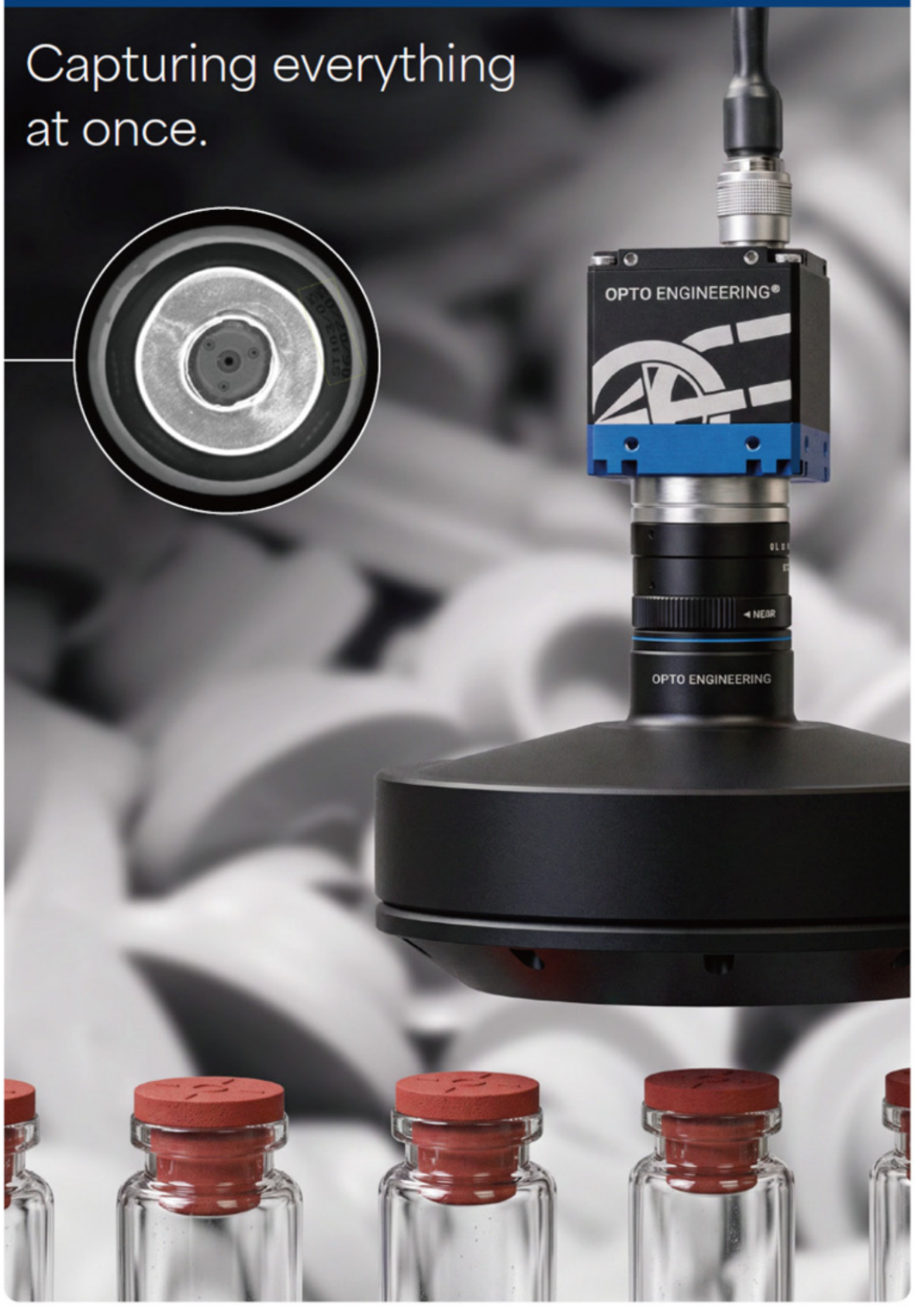
OPTICAL BENCHES

Telecentric optical benches for precision measurements



- Pre-assembled and pre-aligned setup
- Highest accuracy in the telecentric range

360° VIEW OPTICS



360° VIEW OPTICS • Wide range

OUTER INSPECTION



PATENTED

- Complete external inspection in one image
- OCR, code reading & defect detection

INNER INSPECTION

EXTERNAL VIEWING



PATENTED

- Inspect inside without entering
- Complete cavity visibility

INNER INSPECTION

BOROSCOPIC VIEWING



PATENTED

- Inspect where external optics cannot reach
- 360° imaging of hidden features

MULTI-VIEW LENSES

ENTOCENTRIC VIEWING



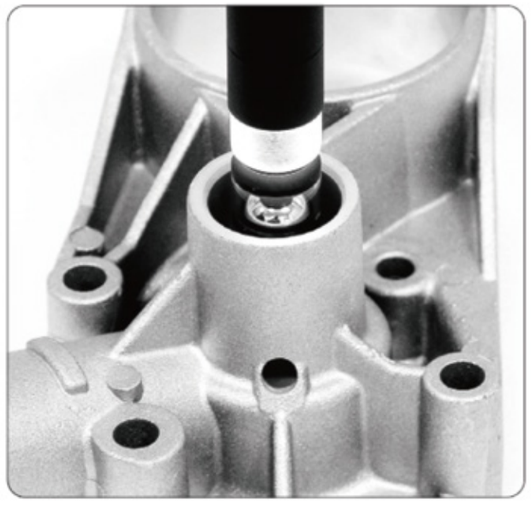
- Multi-view defect inspection in one image
- 45° view for enhanced visibility

MULTI-VIEW LENSES

TELECENTRIC VIEWING



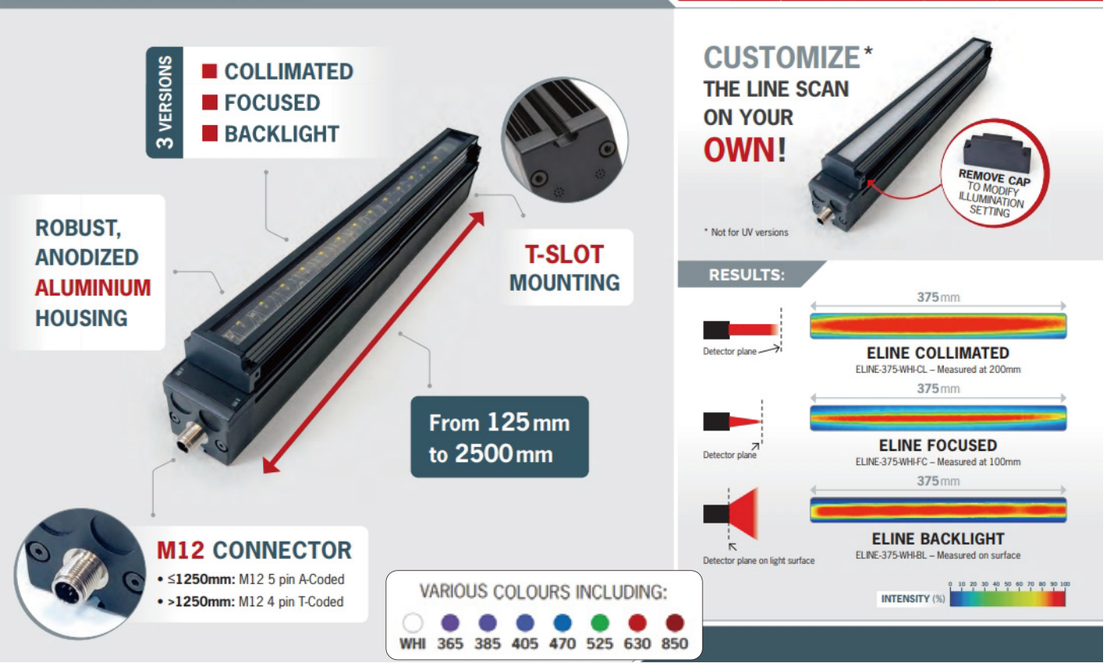
- Multi-view precision measurement in one image
- Consistent magnification across all views



ESSENTIAL **ELINE**

TPL VISION Active LED solutions

FEATURES OVERVIEW



MODULAR **M30 SPOTLIGHT**

TPL VISION Active LED solutions

INTERCHANGEABLE SPOTLIGHT
for Machine Vision Applications





- Flexible, interchangeable spotlights
- Integrated current control
- Autostrobe OverDrive
- 4µs rise time
- IP rating IP67
- Simple M30 mount based on proximity sensor body

VARIOUS COLOURS:

WHI 470 525 630 850 365 385 405

BL-SPOT & IL-SPOT

Z-SPOT

24V ±5% CE UKCA RoHS

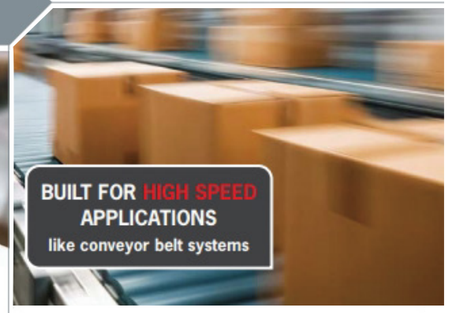
TPL VISION IS AN ISO 9001 CERTIFIED MANUFACTURER.

HIGH POWER STROBE HPS BAR

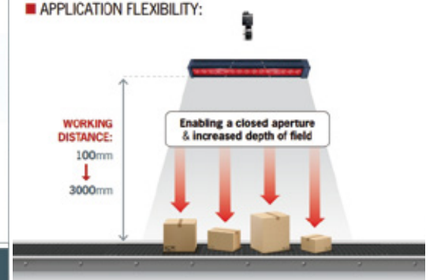
TPL VISION Active LED solutions

WORLD LEADING PERFORMANCE
for demanding Machine Vision & Code Reading Applications





■ APPLICATION FLEXIBILITY:



■ Ensuring shutter speed is short to **remove motion blur** on fast moving objects.

■ **<800ns rise time:** the triggering can be done by the camera for low exposure time applications → **no need for a complex timing controller.**

24V ±5% CE UKCA RoHS

TPL VISION IS AN ISO 9001 CERTIFIED MANUFACTURER.

TPL-CONTROL-1

TPL VISION Active LED solutions

SINGLE-CHANNEL PRECISION LIGHT CONTROLLER
Based on a Modular Approach – Up to 16 Channels



TPL-CONTROL-1
Master & Slave

COMPATIBLE WITH THE EXTERNAL CONTROL (EC) RANGE



TPL-CONTROL-1 is **a precision light controller** with current and voltage regulation to control **any LED illumination setup** in machine vision and image acquisition applications. The lighting can be controlled in a **continuous mode** as well as in a **flash mode**. Currents of **up to 20A** (per channel) are possible. By regulating current and voltage, high efficiency and thus lower heat generation is achieved.

The modular Master/Slave architecture allows the TPL-CONTROL-1 lighting controllers to be used in a wide range of applications.

One TPL-CONTROL-1 module supplies power to one LED light channel. In total, **up to 16 independent lighting systems** can be controlled in this way. Since a **TPL-CONTROL-1 Master** consists of both a communication and a power section, it serves as the configuration interface for the user and all light channels. A **TPL-CONTROL-1 Slave**, on the other hand, has only a power section. All connected channels can be configured conveniently and easily via **a single interface** on the Master module. The respective channel configuration is stored decentrally in each module separately.

ADVANTAGES

- CURRENTS OF UP TO 20A PER CHANNEL
- FLEXIBILITY AND SCALABILITY THROUGH MODULAR DESIGN
- EASY CONFIGURATION VIA INTEGRATED WEBSITE
- ETHERNET AND SERIAL INTERFACE
- WIDE RANGE OF EXTERNAL CONTROL (EC) PRODUCTS



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TPL VISION IS AN ISO 9001 CERTIFIED MANUFACTURER.

주요 레퍼런스 Main Reference

Special sensor Special Sensing & Heavy Industry

ONDOSENSE
SMART SENSOR SOLUTIONS

- 고객사 환경, 장비에 맞춘 제품 선정 및 평가 지원(PoC)

극한 환경에서의 초정밀 레이더 계측 : Ondosense Radar sensor

Target 철강, 금속 주조, 중공업 등 고온 · 분진 · 수증기 상시 발생 공정

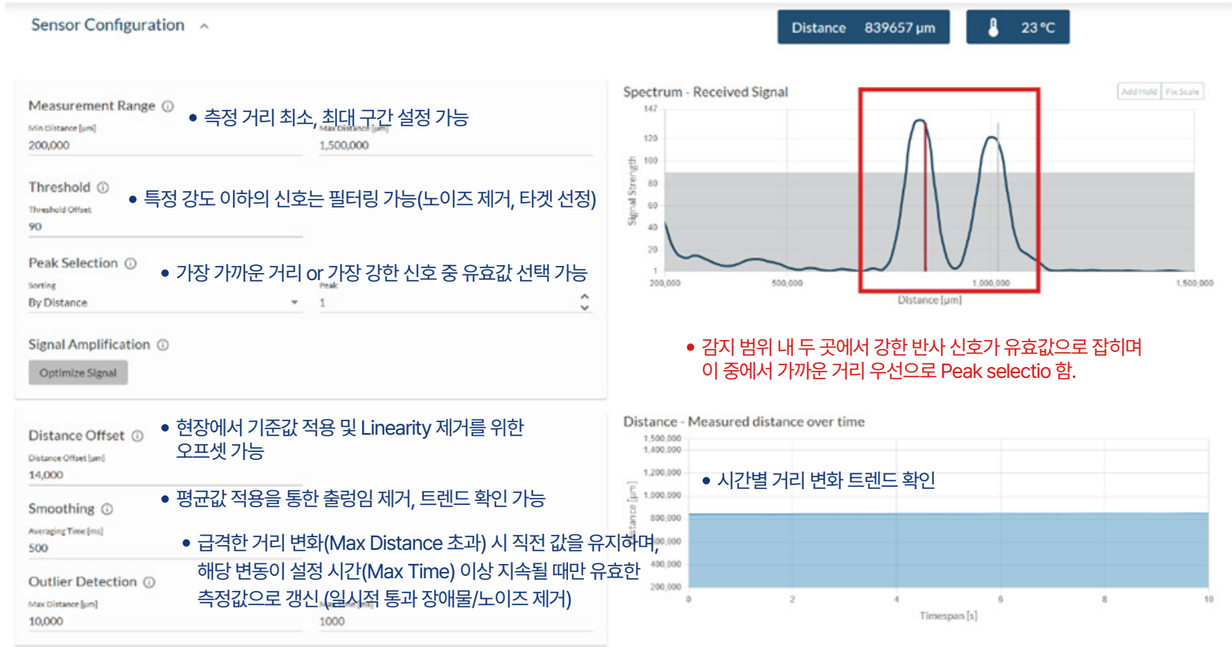
Core Value 레이저, 광학 센서 사용이 불가능한 환경에서 비접촉 마이크로미터(μm)급 거리 · 두께 · 레벨 연속 측정

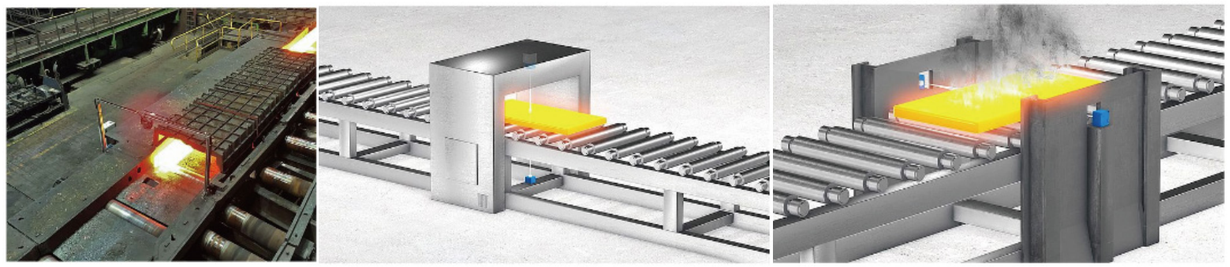
Key Spec & Tech

최대 1μm 정밀도 / 500Hz 고속 응답 레이더 기술
AirChill 하우징 등 열악한 환경 전용 보호 설계

주요 적용 사례

제철소 냉연/열연 강판 폭 · 두께 측정, 쇳물 탕면 레벨 제어, 데스케일러 구간 거리 계측

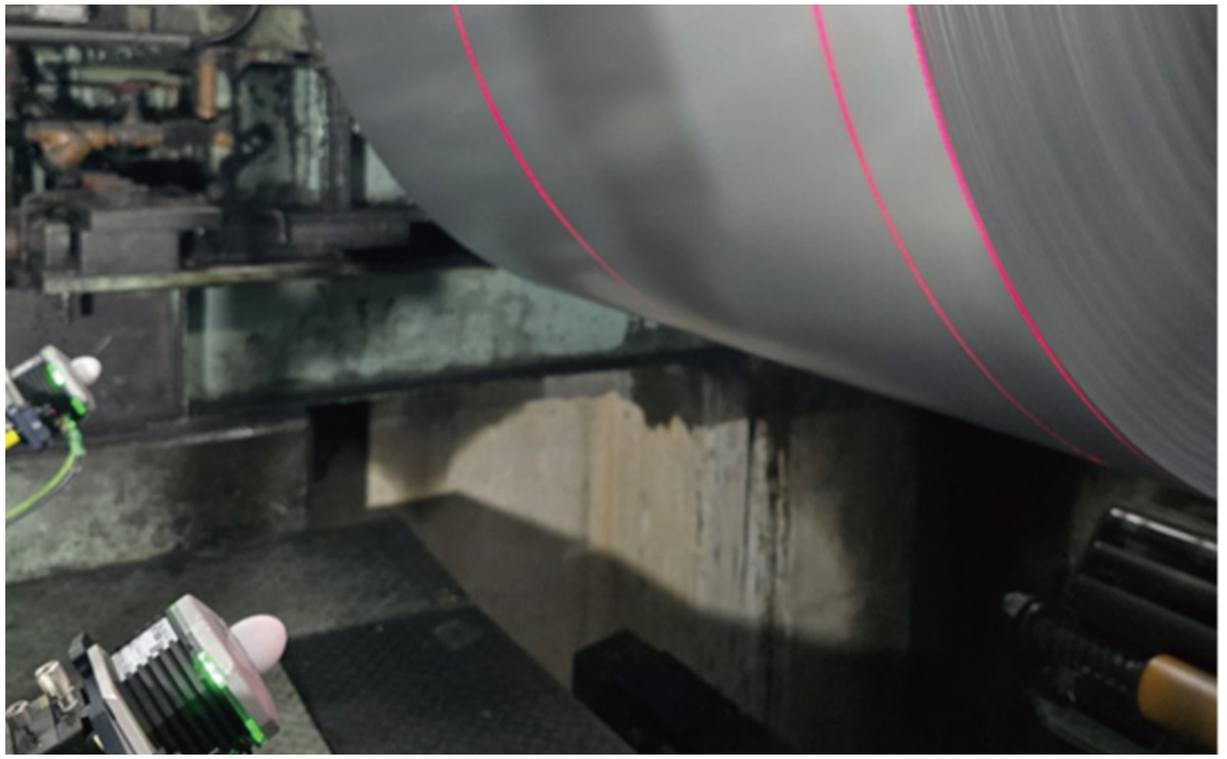




반제품 인라인 치수 측정 현장 사진

슬라브 인라인 두께 측정 (레이더 센서 상하 2개)

슬라브 인라인 폭 측정 (레이더 센서 좌우 2개)



Special sensor **Special Sensing & Heavy Industry**

di-soric

- 고객사 환경, 장비에 맞춘 제품 선정 및 평가 지원(PoC)

고난도 타깃 검출 & 인라인 특화 센서

Target 반도체, 이차전지, 패키징, 정밀 부품 조립 라인

Core Value 일반 센서로 검출하기 까다로운 투명체, 필름 라벨, 메세 단차, 고속 이동체에 대한 맞춤형 신호검출

Key Spec & Tech

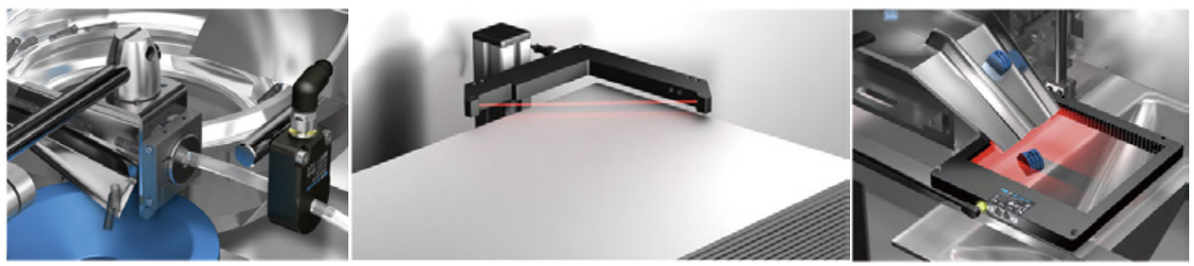
Application-Specific Sensors:
초음파/광학 라벨 센서, 웹 엣지 제어(Web Edge), 미세 색상 차이 판별, 레벨 측정

High-Precision Fork & Optical Sensors
슬롯/포크형 센서를 통한 고속 정밀 위치 결정 및 부품 유무 판별

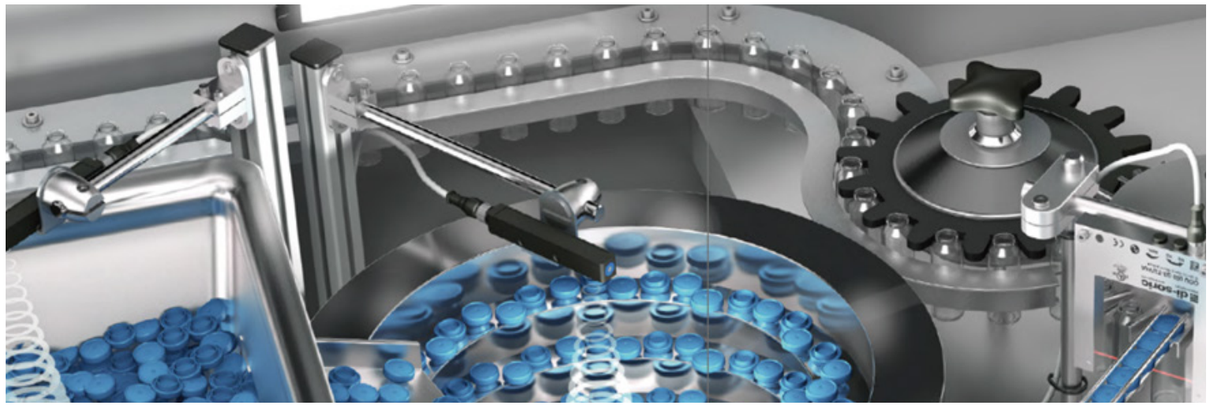
Vision & ID Sensors: 간이 비전 센서 및 컴팩트 산업용 코드 리더기 라인업

주요 적용 사례

이차전지 분리막/필름 사행(Edge) 제어, 제약·패키징 투명 용기/라벨 고속 감지, 정밀 가공품 단차 검출



PART FEEDING
TROUBLE-FREE AND SAFE OPERATION



Our sensors for material feeding

Inductive ring sensors	Inductive tube sensors
Light barriers and diffuse sensors	Our sensors for material feeding
Fork light barriers	Angled light barriers
Frame light barriers	Plastic fiber optic sensors
Glass fiber optic sensors	Ultrasonic barrier/sensors
Vision sensors	

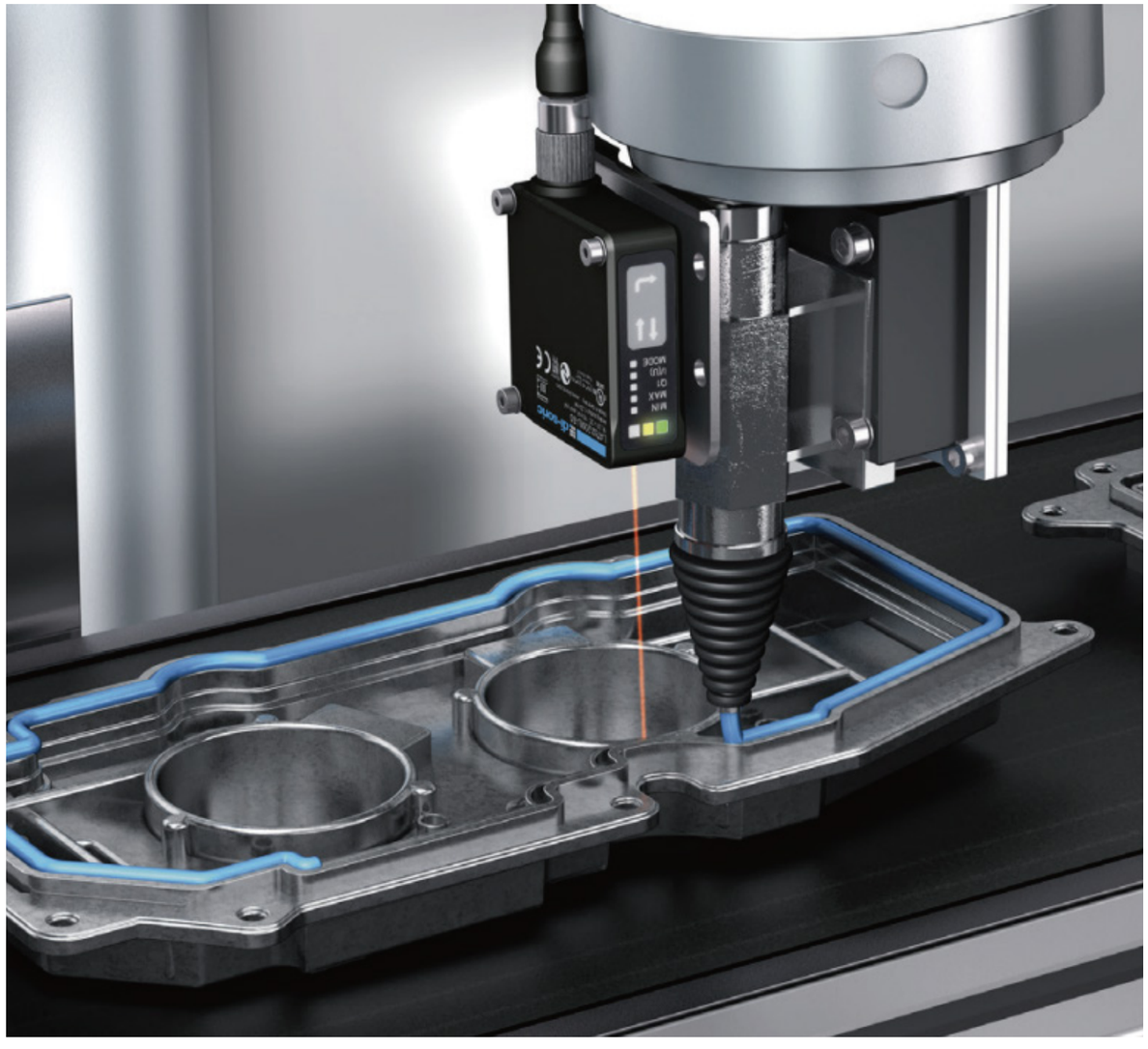
LABELING TECHNOLOGY
WITH PRECISION AND MAXMUM PRODUCTIVITY



Our sensors for material feeding

Inductive proximity sensors	Ultrasonic fork sensors
Light barriers and diffuse sensors	Ultrasonic label sensors
Optical distance sensors	Capacitive label sensors
Plastic fiber optic sensors	Vision sensors
Optical label sensors	

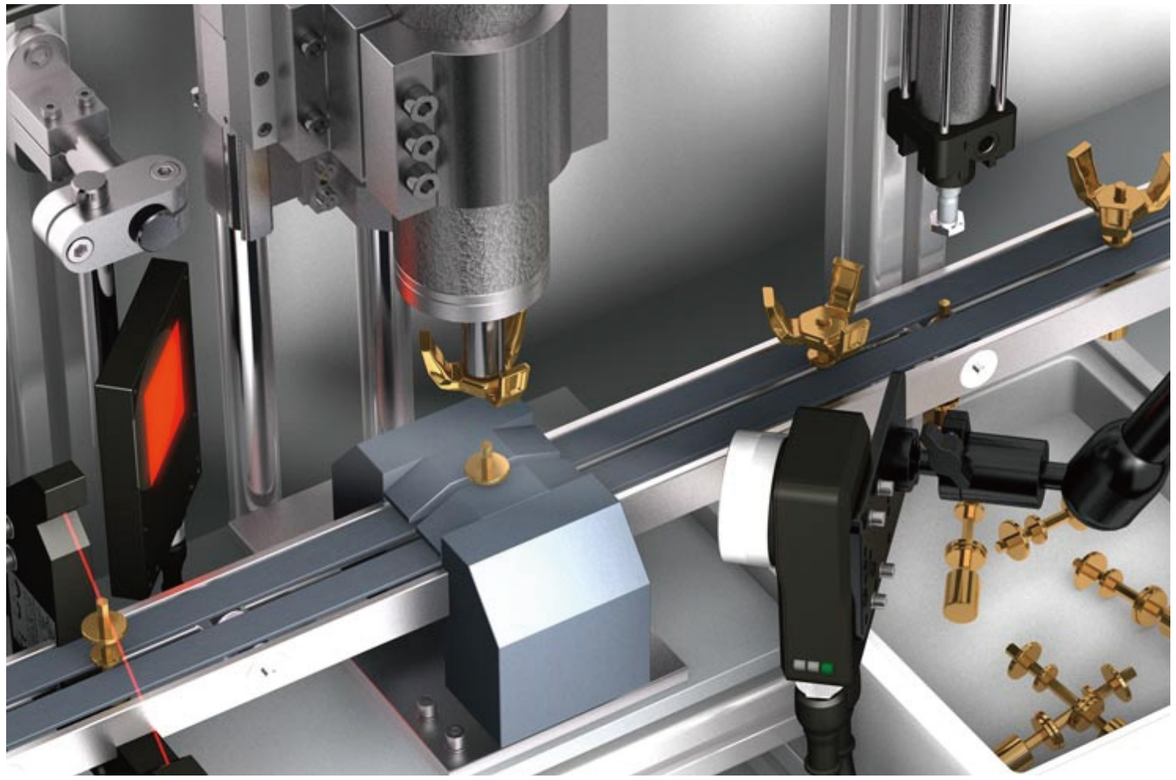
OPTIMAL DISTANCE MEASUREMENT
IN AUTOMATION



Our sensors for material feeding

Optical distances sensors	Ultrasonic barrier/ sensors

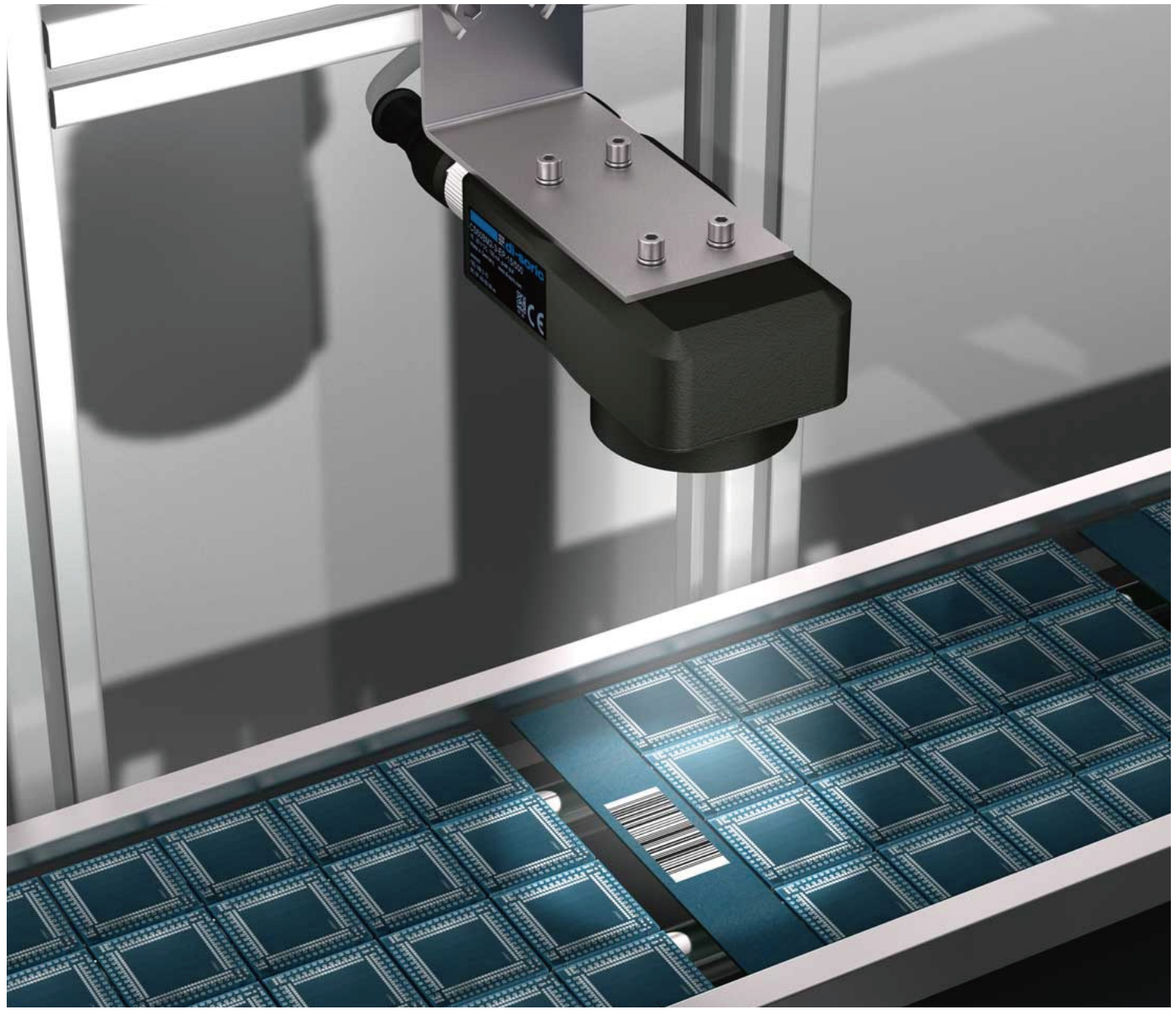
QUALITY INSPECTION
CRUCIAL PROCESS STEP



Our sensors for material feeding

Optical distance sensors	Fixed-mounted ID readers
Color sensors	Vision sensors
Profile sensors	Lighting for industrial image processing
Handheld ID readers	

IDENTIFICATION
CODE READING IN FACTORY AUTOMATION



Our sensors for material feeding

Handheld ID readers	Vision sensors
Fixed-mounted ID readers	

LIGHTING SOLUTIONS
FOR INDUSTRY



Special sensor

YT SOLUTION
Engineering Support

01

현장 맞춤형 센서 Poc 지원

센서 단품 판매에 그치지 않고 현장 환경과
생산 특성을 고려한 최적의 센서 제안 및 테스트,
최적화 지원



Our sensors for material feeding

Machine lighting

Signal lighting



02

PLC/HMI 다이렉트 통신 연동

센서를 상위 제어기와 연동하고 시스템을
구축하기 위한 현장 지원

